

Reflection Essay

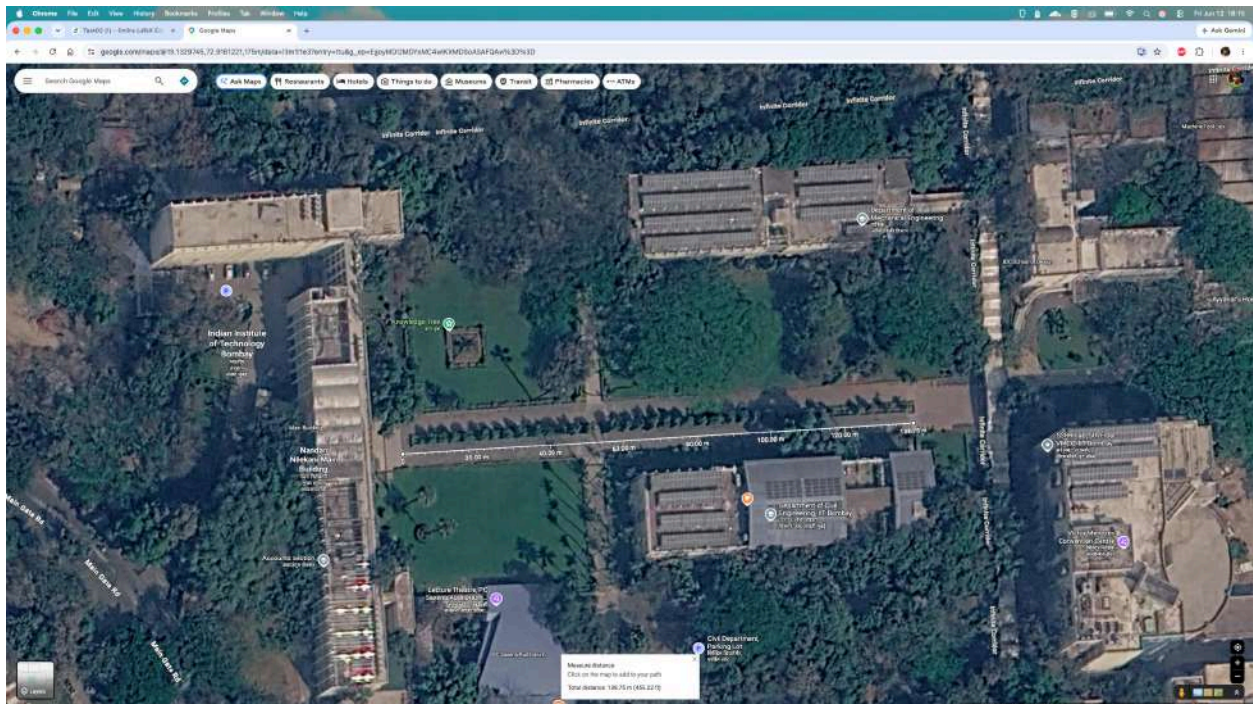
Task 02 | Arya Singh

CV CSO610 | EPGD IIT Bombay

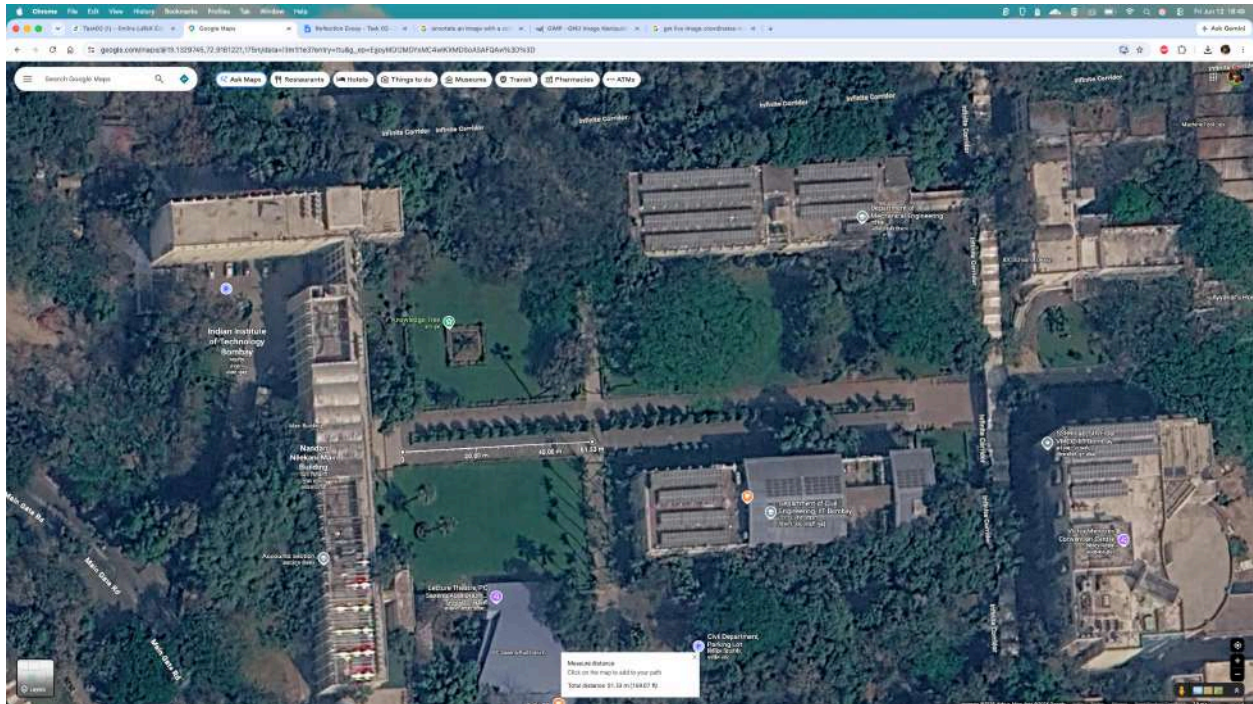
When I first saw this assignment, I thought that this is easier than the previous one as it is mostly just practical with a bit of math. It was not until much later I realized that it had many complexities, many places that I could go wrong, etc. I realized it was quite literally asking us to measure distance using nothing but a single photograph and a bit of projective geometry.

The first part of the assignment used an image of a road on IIT Bombay campus, leading from the Main Building toward the Gyanam-Paramam-Dhyeyam arch. The goal was to find the real-world distance between a parked bicycle and the arch, using only the image and a single known reference distance. Since I don't have access to the campus as of now, I went to Google Maps to find the required distance.

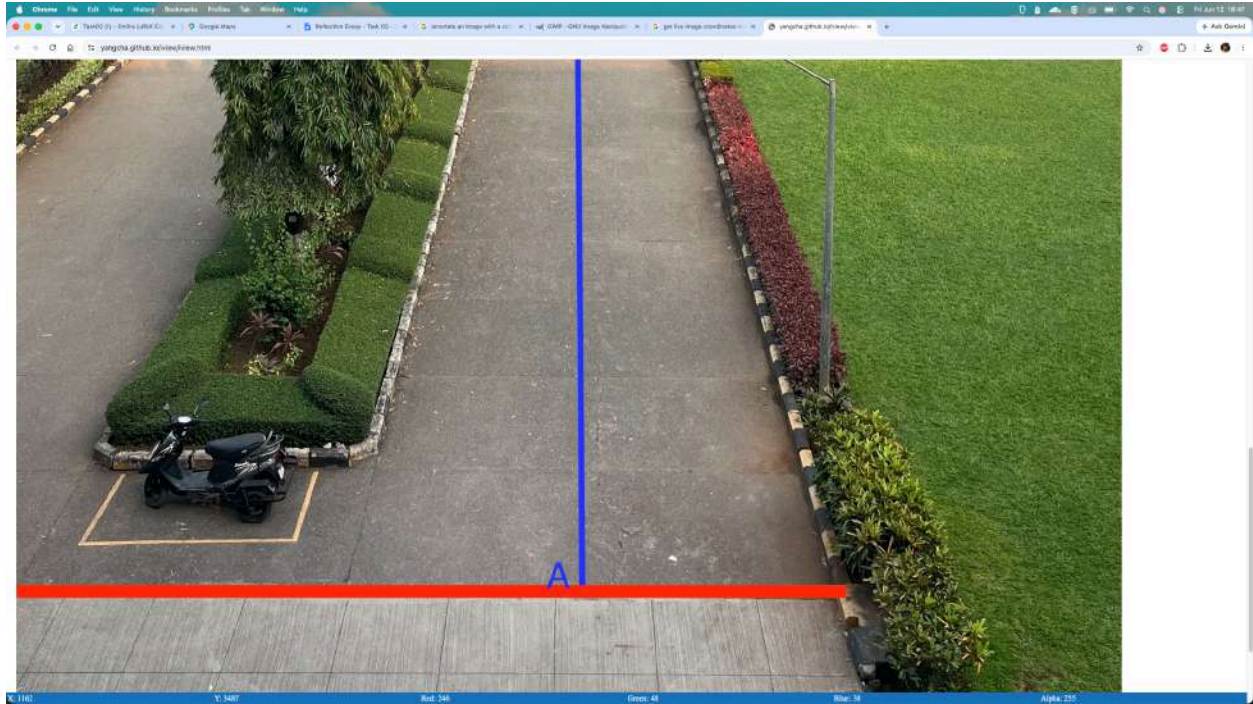
On Google Maps, I had to locate the Main Building and find out which portion of the structure had been captured in the photo provided to us in the zip. This was the first hurdle. Searching for "Gyanam Paramam Dhyeyam arch" returns nothing useful since it's not a named landmark. After some digging I found out this:



I used the image above to find the distance from the 'red line' in the image to the end of the path. It came out to be around 50 meters as seen in the pasted image. Similarly I found some other distances as well, like the distance from red line to the intersection of paths which is pasted below:



I then went on to annotate the image using Preview, the built-in image viewer and editor of macOS. It was fairly intuitive and easy to complete that task. Once I was finished, another hurdle was to find out pixel coordinates. Since I am on a Mac, and don't have MS Paint, I had to find another way. Luckily, I found a utility website which lets me upload an image, and tells me the pixel coordinates as I move my cursor around. The website link and screenshot are attached below. One can notice the X,Y and R,G,B coordinates in the bottom bar.

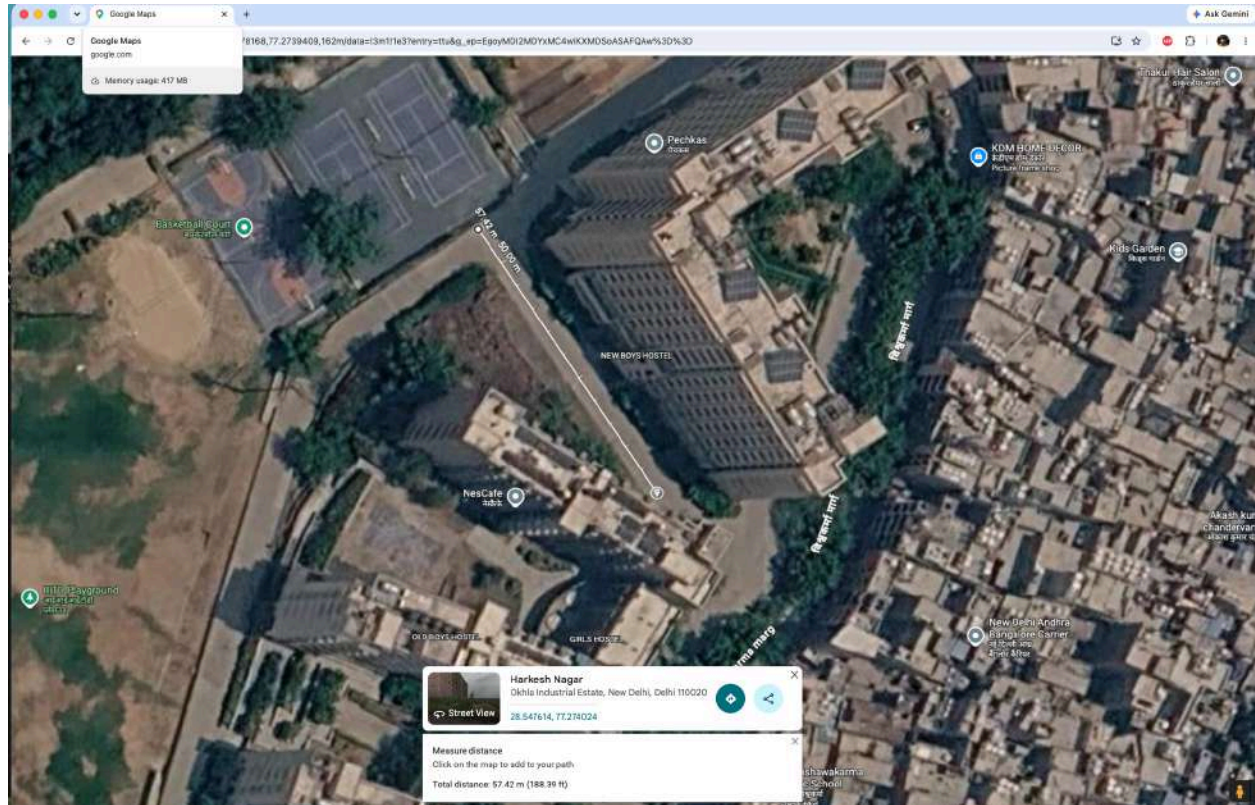


<https://yangcha.github.io/iview/iview.html>

For the object detection portion, I used a pretrained YOLOv8 model to locate the bicycle automatically. I found out that the smallest model (yolov8n) failed to detect the bicycle entirely. It's quite small and partially obscured in the image. Switching to the largest model (yolov8x) and lowering the confidence threshold to 0.01 finally produced a detection. I drew a bounding box using openCV, and that image is included in the submission.

For the second part, I took photos near my hostel in IIIT Delhi campus. I found a long, straight road near my residence with excellent perspective convergence. It had visible curbs running the length of the road, a street light, and a tennis court visible in the distance, so it turned out to be a perfect natural landmark. I took the photo in the early evening, with relatively flat, even lighting, so that it doesn't have harsh shadows, which I suspect helped YOLO's detection accuracy considerably. Since I am doing the task individually, I convinced two of my friends to help me with this. One of them stood on the road at a position I had independently measured (using Google Maps again) to be 13 m from where I was standing with the camera. This measured value served as my ground truth.

The other friend helped capture the third person view. In that photo, both the person and I are visible, with the distant tennis court. To find the distance of the tennis court from us, I again used Google Maps' measure distance feature, shown below:



The cross-ratio prediction came out to 10.73 m, against a measured ground truth of 13 m, which is an error of about 2.27 m, or roughly 17.5%. Given that the Google Maps measurement likely carries an uncertainty of a few meters, and that pixel-level annotation errors near the vanishing point amplify dramatically (as I'd already noticed in the IITB campus question), this felt like a genuine result.

From this assignment, I have learnt a lot about what happens in the real-world when dealing with distances in photos. What struck me most throughout this assignment was how much real-world geometry depends on assumptions that are easy to state but hard to verify: flat ground, perfectly collinear points, an accurately estimated vanishing point. I hope to learn much more practical CV knowledge in the coming tasks.